CYBERTWIN AI

Observe traffic. Explain threats. Test defenses.

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SIMPLE SYSTEM ARCHITECTURE

1 OBSERVE
PCAP + network logs

→

2 UNDERSTAND
IP • ports • protocol

→

3 DETECT
IDS rules + ML

→

4 DECIDE
Review + safe action

→

5 VISUALIZE
Network + reports

ONE CONNECTED WORKFLOW • Offline replay is labeled • Administrator approval controls prevention

WHY CYBERTWIN

See traffic.
Explain alerts.
Test defenses.

Computer Networks + AI

NETWORK INPUTS

PCAP / PCAPNG
Offline packet captures

SURICATA / ZEEK
Connection and alert logs

SIMULATION
Reproducible local examples

Provenance always visible:
SIMULATION · REPLAY · LIVE

NETWORKING CONCEPTS

01 **IP addresses**
Who communicated?

02 **Ports + protocol**
Which service and transport?

03 **Packets + bytes**
How much was observed?

PRESENTATION DEMO

1. Upload sample capture
2. Inspect the scan alert
3. Approve a simulated block
4. Compare virtual defenses

React · FastAPI · SQLite
Scapy · NetworkX · scikit-learn

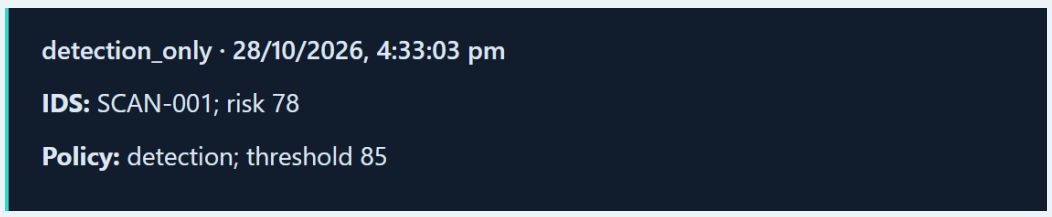
PROJECT IN ACTION | ACTUAL SCREENSHOTS

A. What traffic was observed?



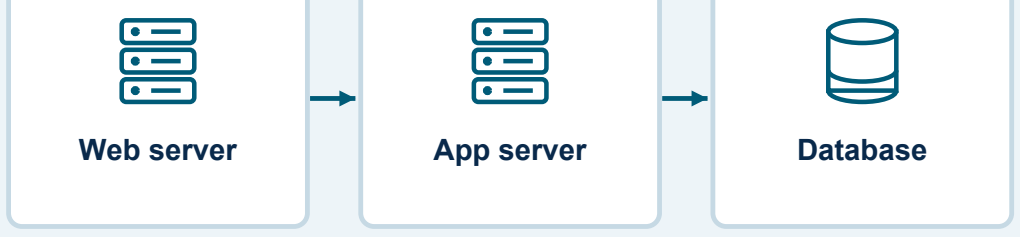
Owned offline capture • counters from stored flows

B. Why did the IDS flag it?



Port-scan evidence + rule risk + IPS policy in one view.

DIGITAL TWIN | A SIMPLE DEFENSE EXAMPLE



BEFORE Direct web → database access

AFTER Restrict the direct link; use the app server

Modeled access only. No physical route or real router change is claimed.

REPLAY RESULTS YOU CAN VERIFY

13	11	1
PACKETS	CONVERSATIONS	IDS ALERT

Normal TCP, DNS/UDP and ICMPv6 + an owned synthetic scan example.
No traffic was sent to external targets.

TESTED LOCALLY

71 backend tests

11 frontend tests

10 browser checkpoints

SUPPORTING AI

Actual confusion matrix



0.757
Synthetic test-set F1
1,800 train / 600 test rows
Supports IDS; does not decide IPS.

SAFE PREVENTION

SIMULATED

No real packets blocked

Approval, expiry and action history are recorded.

CONCLUSION + NEXT

One connected view: traffic → evidence → decision → visual analysis.

Next: isolated Linux enforcement validation and real-data evaluation.